RAGNexus - Hybrid RAG Chatbot

-Vedant Shimpi

1. Project Overview & Chatbot Objective

RAGNexus is a full-stack, AI-powered chatbot built on a Retrieval-Augmented Generation (RAG) architecture. Unlike traditional chatbots that reply from a fixed script or a generic language model, RAGNexus grounds every answer in documents that the user explicitly provides URLs, PDFs, TXT files, or CSVs. This ensures responses are factual, traceable, and accompanied by source citations.

The system combines a FastAPI backend, a React + Vite frontend, a Qdrant vector database for semantic search, a BM25 keyword retriever, a cross-encoder reranker, and the Groq LLM (llama-3.1-8b-instant) for answer generation. When no document is loaded, the chatbot falls back to a general-purpose conversational mode.

Core Objective:

- Allow users to upload or link any document (PDF, URL, TXT, CSV)
- Answer natural-language questions strictly from ingested content
- Return answers with cited source chunks for full transparency
- Maintain persistent chat sessions and conversation history
- Provide a general fallback chat mode when no document is loaded

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2. Target Users & Use Case

RAGNexus is designed as a general-purpose intelligent document assistant. Its primary target audiences are:

Students & Researchers Upload research papers, textbooks, or lecture notes and ask direct questions without reading the entire document. Ideal for summarising key findings or cross-referencing multiple sources.

Customer Support Teams Ingest product manuals, FAQs, or policy documents. The chatbot acts as a first-line support agent that retrieves precise answers from company documentation.

Legal & Compliance Professionals Upload contracts, regulations, or guidelines and query them for specific clauses, obligations, or definitions.

Campus Assistants Ingest handbooks, timetables, or notice boards and allow students to query course schedules, hostel rules, exam dates, and more.

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3. System Architecture

RAGNexus is structured around two main data flows:

Document Ingestion Flow:

Source (URL / PDF / TXT / CSV)
→ Loader (fetches & parses content)
→ Chunker (splits into overlapping chunks)
→ Embedder (sentence-transformers model)
→ PostgreSQL (document & chunk metadata storage)
→ Qdrant (dense vector storage)

Question-Answering Flow:

User Question
→ HyDE Expansion (generate hypothetical answer for better embedding)
→ Hybrid Retriever (Qdrant dense search + BM25 keyword search)
→ Cross-Encoder Reranker (re-scores retrieved chunks)
→ Top-K Contexts selected
→ Groq LLM (llama-3.1-8b-instant) generates grounded answer
→ Answer + Source Citations returned to frontend

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4. Conversation Flow Design

The chatbot follows a structured conversation flow across two modes: Document Mode (RAG-grounded) and Basic Chat Mode (general LLM).

Step 1 — Welcome & Onboarding The frontend greets the user, presents the option to upload a document or start a chat session, and creates a unique session ID stored in PostgreSQL.

Step 2 — Document Ingestion (optional) The user pastes a URL or uploads a file. The backend ingests, chunks, embeds, and confirms success with a document_id. The UI displays the document in the workspace.

Step 3 — User Query The user types a natural-language question in the chat input. The frontend sends the question along with the session_id and (optionally) document_id to the /ask endpoint.

Step 4 — Retrieval & Answer Generation The backend runs the HyDE → hybrid retrieval → rerank → Groq generation pipeline and returns a grounded answer with numbered source citations.

Step 5 — Follow-up & History The conversation continues with full history persisted per session. The user can ask follow-up questions referencing prior context.

Step 6 — Fallback (No Document) If no document_id is provided or the query falls outside retrieved context, the system routes to /chat/basic which calls Groq directly without retrieval constraints.

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5. Fallback & Error Handling

RAGNexus implements multi-level fallback and error handling to ensure the chatbot always responds gracefully.

No Document Loaded When a user asks a question without providing a document_id, the request is automatically routed to POST /chat/basic, which calls Groq directly for a general answer.

Context Not Found in Document If the hybrid retriever returns no relevant chunks above the confidence threshold, the system responds: *"I*

could not find relevant information in the provided document for your question. Please try rephrasing, or use basic chat mode for general queries."

Invalid File Type The /ingest/file endpoint validates file extensions. If an unsupported type is uploaded, the API returns HTTP 422 with the message: *"Only PDF, TXT, and CSV files are supported."*

Qdrant / Database Unavailable If the vector store or PostgreSQL is unreachable, the backend returns HTTP 503 with a descriptive error message and logs the exception for debugging.

Malformed / Empty Input Empty queries or missing required fields trigger Pydantic validation errors returned as HTTP 422 with field-level error descriptions.

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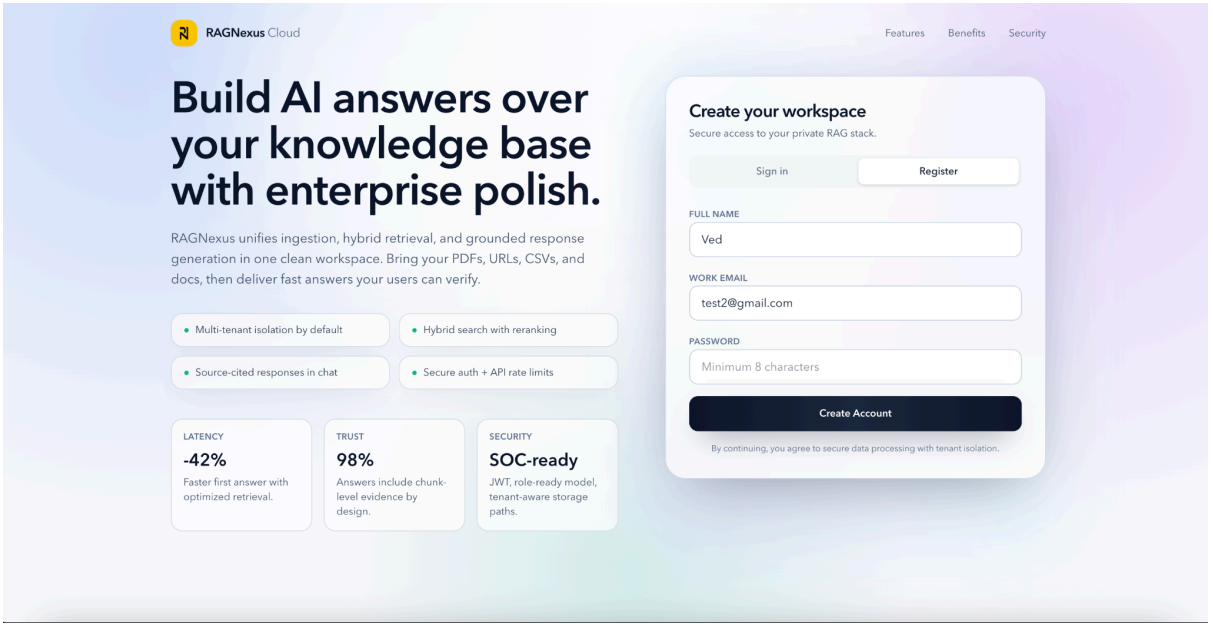
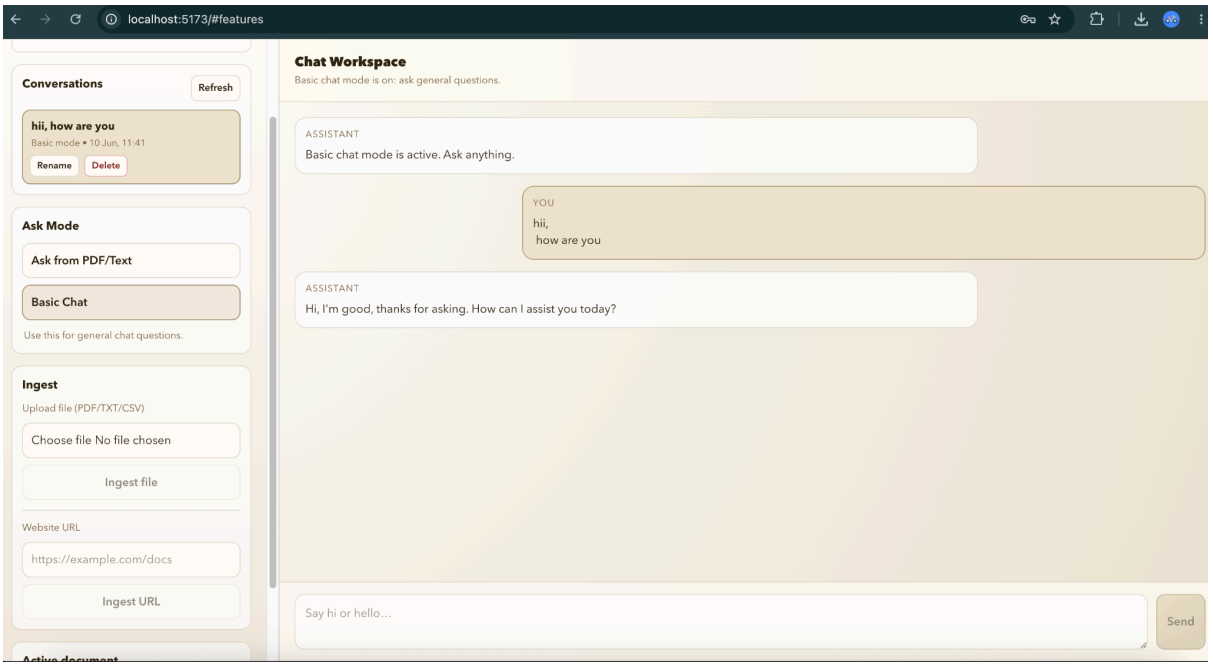
6. Tech Stack & Setup

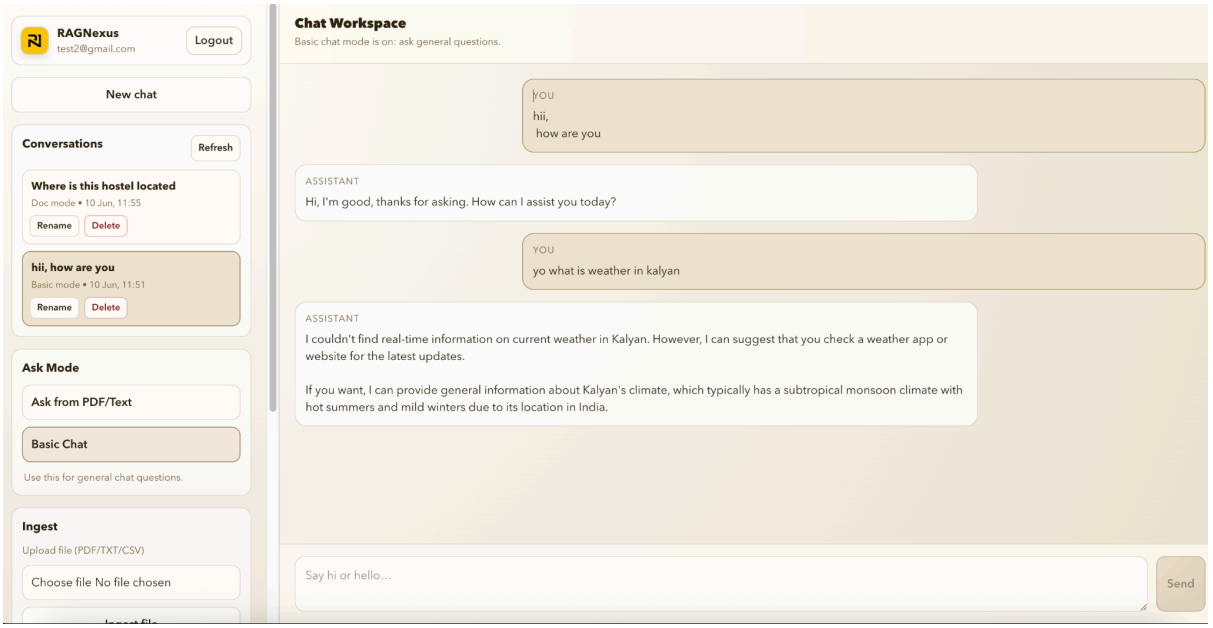
Component	Technology
Backend Framework	FastAPI + SQLAlchemy (Python 3.10+)
Frontend	React + Vite (Node.js 18+)
Vector Database	Qdrant (Docker)
Relational Database	PostgreSQL / Supabase
LLM Provider	Groq (llama-3.1-8b-instant)
Embeddings	sentence-transformers (all-MiniLM-L6-v2)
Keyword Retrieval	BM25 (rank_bm25)
Reranker	Cross-Encoder (sentence-transformers)

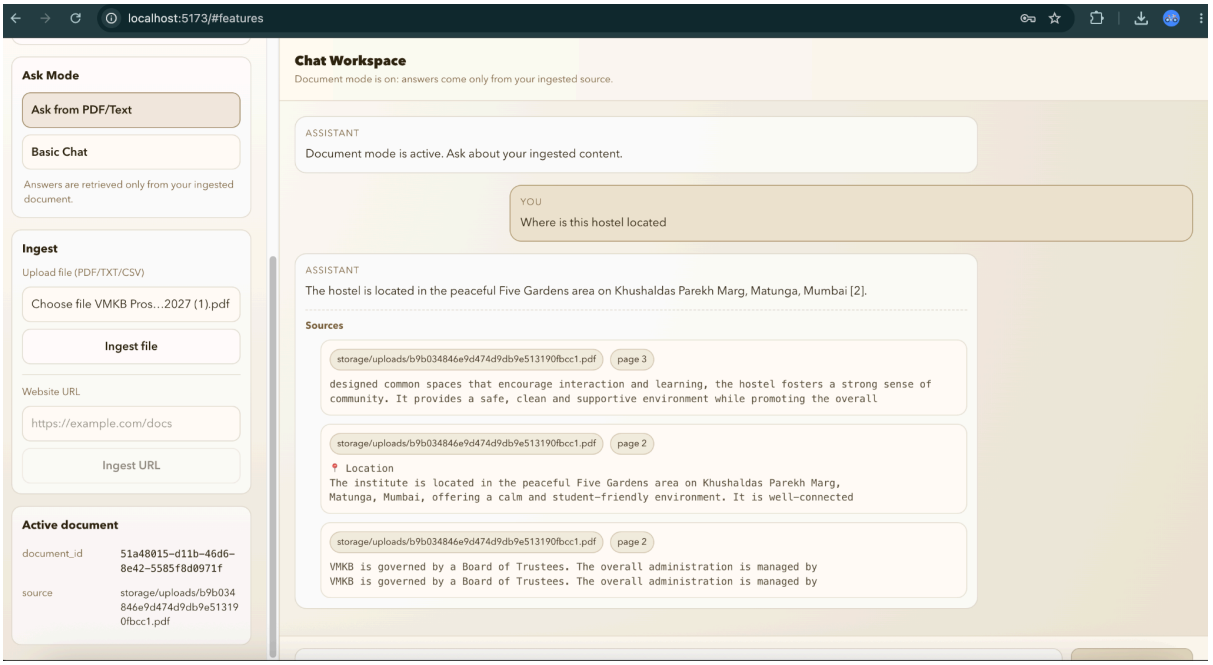
Evaluation (optional)	RAGAs

Screenshots :






RAGNexus
test2@gmail.com
Logout
New chat
Conversations
Refresh
Where is this hostel located
Doc mode • 10 Jun, 11:55
Rename
Delete
hii, how are you
Basic mode • 10 Jun, 11:51
Rename
Delete
Ask Mode
Ask from PDF/Text
Basic Chat
Use this for general chat questions.
Ingest
Upload file (PDF/TXT/CSV)
Choose file No file chosen
Chat Workspace
Basic chat mode is on: ask general questions.
YOU
hii,
how are you
ASSISTANT
Hi, I'm good, thanks for asking. How can I assist you today?
YOU
yo what is weather in kalyan
ASSISTANT
I couldn't find real-time information on current weather in Kalyan. However, I can suggest that you check a weather app or website for the latest updates.
If you want, I can provide general information about Kalyan's climate, which typically has a subtropical monsoon climate with hot summers and mild winters due to its location in India.
Say hi or hello…
Send


localhost:5173/#features
Ask Mode
Ask from PDF/Text
Basic Chat
Answers are retrieved only from your ingested document.
Ingest
Upload file (PDF/TXT/CSV)
Choose file VMKB Pros…2027 (1).pdf
Ingest file
Website URL
https://example.com/docs
Ingest URL
Active document
document_id
51a48015-d11b-46d6-8e42-5585f8d0971f
source
storage/uploads/b9b034846e9d474d9db9e513190fbcc1.pdf
Chat Workspace
Document mode is on: answers come only from your ingested source.
ASSISTANT
Document mode is active. Ask about your ingested content.
YOU
Where is this hostel located
ASSISTANT
The hostel is located in the peaceful Five Gardens area on Khushaldas Parekh Marg, Matunga, Mumbai [2].
Sources
storage/uploads/b9b034846e9d474d9db9e513190fbcc1.pdf
page 3
designed common spaces that encourage interaction and learning, the hostel fosters a strong sense of community. It provides a safe, clean and supportive environment while promoting the overall
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page 2
Location
The institute is located in the peaceful Five Gardens area on Khushaldas Parekh Marg, Matunga, Mumbai, offering a calm and student-friendly environment. It is well-connected
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page 2
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RAGNexus
test2@gmail.com
Logout
New chat
Conversations
Refresh
Where is this hostel located
Doc mode • 10 Jun, 11:55
Rename
Delete
hii, how are you
Basic mode • 10 Jun, 11:51
Rename
Delete
Ask Mode
Ask from PDF/Text
Basic Chat
Answers are retrieved only from your ingested document.
Ingest
Upload file (PDF/TXT/CSV)
Choose file No file chosen
YOU
what is eligibility to get admission in this hostel
ASSISTANT
The eligibility to get admission in this hostel is the following:
* Candidates who have completed a course of full time training which must be recognized by the government or the University of Mumbai [3]
* Candidates who have filled in the prescribed application form for admission and submitted the required documents [1]
Sources
storage/uploads/b9b034846e9d474d9db9e513190fbcc1.pdf
page 4
with the required documents.
Download Hostel Admission Application Form
After interview, the admission may be granted for that academic year only at the
discretion of the Admission Committee.
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page 8
in dismissal.
4. Students leaving the hostel for vacations, personal reasons or family functions must
obtain prior written permission.
storage/uploads/b9b034846e9d474d9db9e513190fbcc1.pdf
page 4
of full time training which must be recognized by the government or the University
of Mumbai.
4.Who has filled in the prescribed application form for admission and submitted the